

Tuning page allocation

Here's how you can check and modify the current page size:

```
[root@legacy ~]# cat /proc/sys/vm/min_free_kbytes
3031
[root@legacy ~]#
[root@legacy ~]# echo 4000 > /proc/sys/vm/min_free_kbytes
[root@legacy ~]# cat /proc/sys/vm/min_free_kbytes
4000
[root@legacy ~]#
```

To make this change permanent across reboots, append the following as in your `/etc/sysctl.conf` file:

```
vm.min_free_kbytes = 4000
```

...and run `sysctl -p`. Please note that it should only be configured when an application regularly needs to allocate large blocks of memory and then frees up the same memory.

Memory overcommit

Airlines sometimes tend to sell more tickets than the actual number of seats available. This is to be on the safe side in case a number of passengers cancel their tickets at the last moment. Linux manages memory in a very similar fashion. Whenever any process asks for memory allocation, Linux allocates it assuming that a program is asking for more memory than it actually requires.

Memory overcommit can be set by defining the `vm.overcommit_memory` parameter under the `/etc/sysctl.conf` file. It takes three values:

- 0 – means heuristic overcommit (i.e., always refuses to over-commit)
- 1 – means always overcommit (whether available or not)
- 2 – commits all of swap plus a percentage of RAM, controlled by `vm.overcommit_ratio` (default is 50).

We can verify the current values of the overcommit parameters as follows:

```
[root@legacy ~]# sysctl -a | grep overcommit
vm.overcommit_ratio = 50
vm.overcommit_memory = 0
```

In order to change the parameter from 0 to 1, we can simply modify the parameter in the `/etc/sysctl.conf` file as follows:

```
vm.overcommit_memory = 1
```

If the `vm.overcommit_memory` is set to 2, then the system will allocate memory equal to the amount of swap plus a percentage (default 50 per cent) of

physical memory. We can control the `vm.overcommit_ratio` also by appending a corresponding entry in `/etc/sysctl.conf` as follows:

```
vm.overcommit_ratio = 40
```

As always, do run `sysctl -p` to commit your changes.

ARP cache

Address Resolution Protocol (ARP) resolves or maps IP addresses to hardware addresses (MAC). In Linux, by default, the ARP cache size is limited to 512 entries as the soft limit and 1,024 entries as hard limit. The soft limit automatically becomes the hard limit after five seconds.

To see the current soft and hard limits, you can use `sysctl -a | grep -i thresh`, as shown:

```
[root@legacy ~]# sysctl -a | grep -i thresh
kernel.sched_shares_thresh = 4
kernel.random_read_wakeup_threshold = 64
kernel.random_write_wakeup_threshold = 128
kernel.softlockup_thresh = 60
net.netfilter.nf_conntrack_frag6_low_thresh = 196608
net.netfilter.nf_conntrack_frag6_high_thresh = 262144
net.ipv4.route_gc_thresh = 32768
net.ipv4.neigh.default_gc_thresh1 = 128
net.ipv4.neigh.default_gc_thresh2 = 512
net.ipv4.neigh.default_gc_thresh3 = 1024
<<<=output truncated>>>
```

We can easily configure these values under the `/etc/sysctl.conf` file as explained earlier. One very interesting attribute here is the *garbage collection frequency* (time in seconds). When the hard limit is exceeded, the kernel garbage collector rescans the cache after the time specified and purges entries to take it under the limit.

```
[root@legacy ~]# sysctl -a | grep gc_interval
net.ipv4.route_gc_interval = 60
net.ipv4.neigh.default_gc_interval = 30
net.ipv6.neigh.default_gc_interval = 30
net.ipv6.route_gc_interval = 30
```

I'll end here, for this month. In the next issue I will conclude this series by explaining memory reclamation and network performance boosting. 

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